

Lab04

This lab has three parts:

- **Part 1** walks you through examples on how to use R to run nonparametric two-sample t-tests (Wilcoxon rank sum test and the Sign-test)
- **Part 2** illustrates how to use R to compare proportions
- **Part 3** provides self-assessment opportunities for the two topics in this lab.

#0: Reading an excel spreadsheet:

This is just a refresher of the explanations provided in the R introduction manual, section 11.6. When you have a .xls or .xlsx file, there are at least two ways you could read this into R. One is to store a specific spreadsheet as a comma delimited file (.csv format) prior to reading it in R. The other is to read the Excel workbook (.xlsx format file). The ability to read an Excel workbook is not part of base R. It is added to R by functions in the library `readxl`. This is the one I find most useful. You are welcome to use a different one.

```
# read in the data
library(readxl) # make sure to install if it isn't already installed
hb <- read_excel('hamburger.xlsx')
head(hb)
```

```
# A tibble: 6 x 2
  treatment    cfu
  <chr>      <dbl>
1 control    0.17
2 control    0.35
3 control    0.22
4 control    2.5
5 control    0.53
6 control    0.57
```

The function `read_excel()` in the `readxl` library will read both `.xlsx` and `.xls` format files. Just specify the appropriate filename.

The result from `'read_excel()'` is a tibble. This is an extension of the data frame that we use to store data. You can treat the tibble just like a data frame. The biggest difference is that a large tibble is printed more nicely than a large data frame.

#1. Wilcoxon ranks-sum test

The syntax is exactly like `t.test`: a formula with the response variable (here, `cfu`) $\sim$ the treatment variable (here, `treatment`), and the name of the data frame.

```
wilcox.test(cfu~treatment, data=hb, correct=F)
wilcox.test(cfu~treatment, data=hb, exact=T)
```

Wilcoxon rank sum test

```
data: cfu by treatment
W = 2, p-value = 0.01027
alternative hypothesis: true location shift is not equal to 0
```

Wilcoxon rank sum test with continuity correction

```
data: cfu by treatment
W = 2, p-value = 0.01291
alternative hypothesis: true location shift is not equal to 0
```

Notes:

- I prefer to not use a continuity correction, so I add `correct=F`. If the data set is small, I prefer the permutation test p -value, which you get by adding `exact=T`, as in the second line.
- One slightly irritating limitation: the algorithm used by R to get the exact p -value does not work when there are tied values (two observations with the same response value). R will print a warning when it can't compute the exact p -value. The exact p -value is well defined, even when there are ties. It's just that R won't calculate it.

If you needed to check calculations by following the steps in the test, you could use R to calculate the ranks and their sums. Please note that you need to make sure you are adding up separately the ranks for each of the groups, in this case control and active.

```
hb$Ranks<- rank(hb$cfu, ties.method = "average")

sum(hb$Ranks[hb$treatment=="control"])
sum(hb$Ranks[hb$treatment=="active"])
```

```
[1] 55
[1] 23
```

To conduct a non-parametric test for paired data (we will use here the schizophrenia example again), we use the same R function, but this time we choose to specify that data are paired (`pair=T`):

```
##Segment 8
schiz <- read.table('case0202.txt', header=T, as.is=T)

wilcox.test(schiz$unaff, schiz$aff, paired = T)
```

Wilcoxon signed rank exact test

```
data: schiz$unaff and schiz$aff
V = 111, p-value = 0.002014
alternative hypothesis: true location shift is not equal to 0
```

Finally, you can use the `SignTest()` function to conduct the sign-test in R, which is implemented in the `DescTools` package:

```
library(DescTools)

SignTest(schiz$unaff, schiz$aff)
```

Dependent-samples Sign-Test

```
data: schiz$unaff and schiz$aff
S = 14, number of differences = 15, p-value = 0.0009766
alternative hypothesis: true median difference is not equal to 0
96.5 percent confidence interval:
 0.04 0.40
sample estimates:
median of the differences
      0.11
```

#2. Comparing proportions

This section provides explanation of the steps in R to conduct the analyses described in Part 3 of Module 2 (compare proportions.)

Since this part of the material is centered around categorical data, we will make use of the R function `factor()`.

Many times, we want to have the categories sorted in a specific way. In R, by default, categories are ordered alphabetically. You can change the order of categories by using the function `factor()`. For example, for the example on Eye and Hair color, we could re-arrange the categories as:

```
##Segment 9

eh.colors<- read.csv("m2_s9.hairandeyecolor.csv", header = T)
eh.colors$EyeColor<- factor(eh.colors$EyeColor,
                             levels = c("Blue", "Hazel/Green", "Brown", "Other"))

eh.colors$HairColor<- factor(eh.colors$HairColor,
                              levels<- c("Blonde", "Red", "Brown", "Black", "Other"))
```

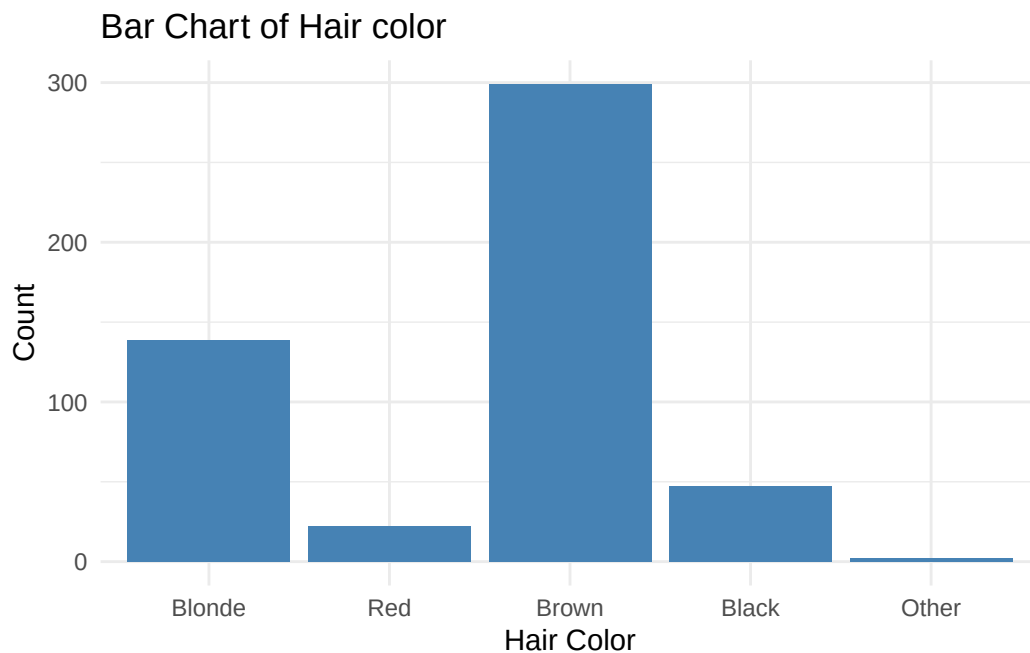
For any categorical variable, you can obtain a frequency table by using the `table()` function from the DescTools library. This function will return a 2-dimensional table and can include, by request, its relative frequencies, a short summary containing the total cases, the dimensions of the table, chi-square tests and some association measures as phi-coefficient, contingency coefficient and Cramer's V. Tables with higher dimensions will simply be printed as flat table, with marginal sums for the first and for the last dimension.

```
table(eh.colors$EyeColor, eh.colors$HairColor)
```

	Blonde	Red	Brown	Black	Other
Blue	90	9	96	1	1
Hazel/Green	36	9	83	1	1
Brown	9	3	117	42	0
Other	3	1	3	3	0

We will continue to use the package ggplot2 for graphing. To obtain a bar graph for these data,

```
library(ggplot2)
ggplot(eh.colors, aes(x=`HairColor`))+
  geom_bar(fill = "steelblue") +
  labs(
    title = "Bar Chart of Hair color",
    x = "Hair Color",
    y = "Count"
  ) +
  theme_minimal()
```



To represent the two variables simultaneously, we use the ggplot function in ggplot2, using the mosaic option. If we choose to plot the data for Hair Color given the Eye Color, we specify this in the ggplot function as:

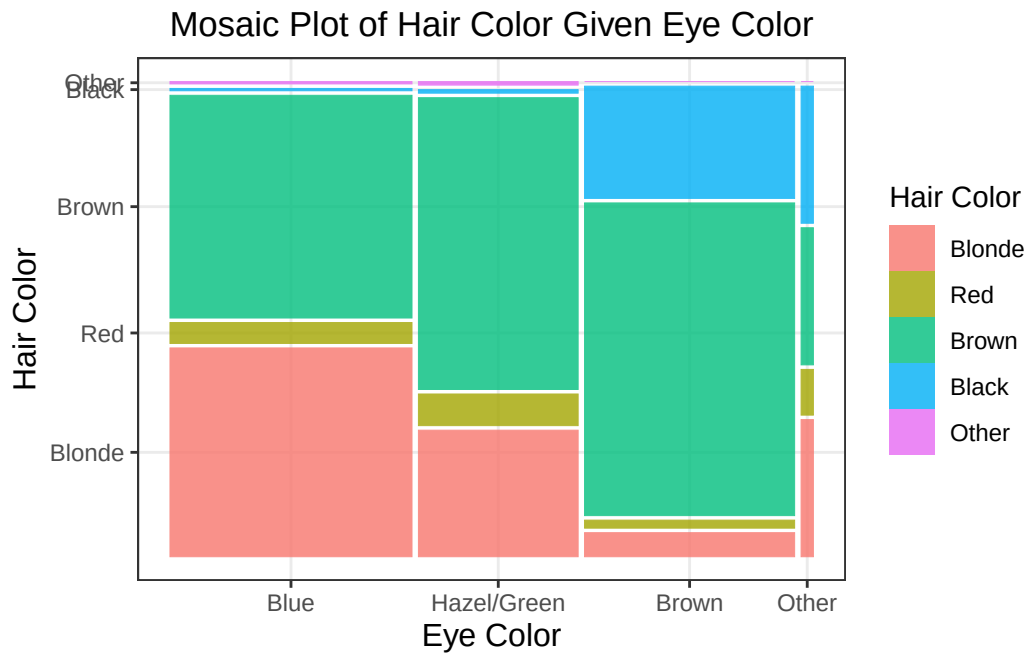
```
library(ggmosaic)

ggplot(data = eh.colors)+
  geom_mosaic(aes(x = product(EyeColor),
                      fill = HairColor), na.rm = TRUE)+
  theme_bw()+
  theme(plot.title = element_text(hjust=0.5,
                                   size = rel(1.2)),
```

```

axis.title.y = element_text(size = rel(1.1)),
axis.title.x = element_text(size = rel(1.1)),
strip.text.y = element_text(size = rel(1.1))) +
labs(x = "Eye Color",
y = "Hair Color",
fill = "Hair Color",
title = "Mosaic Plot of Hair Color Given Eye Color")

```



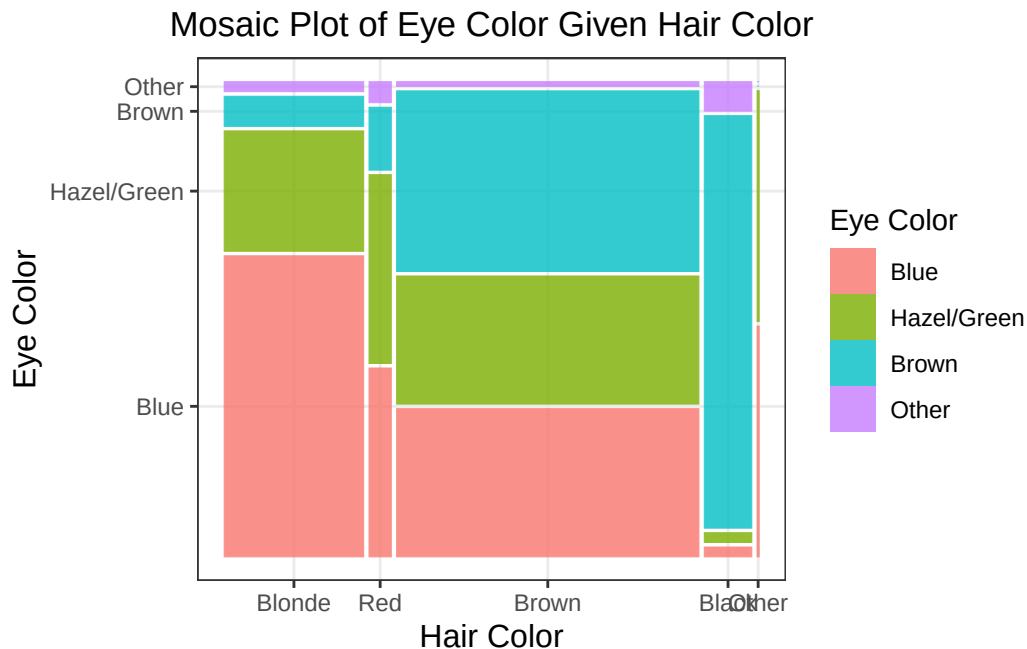
To repeat but this time representing Eye Color given Hair Color, we modify the ggplot option as:

```

ggplot(data = eh.colors) +
  geom_mosaic(aes(x = product(HairColor),
                    fill = EyeColor), na.rm = TRUE) +
  theme_bw() +
  theme(plot.title = element_text(hjust=0.5,
                                   size = rel(1.2)),
        axis.title.y = element_text(size = rel(1.1)),
        axis.title.x = element_text(size = rel(1.1)),
        strip.text.y = element_text(size = rel(1.1))) +
  labs(x = "Hair Color",
y = "Eye Color",

```

```
fill = "Eye Color",
title = "Mosaic Plot of Eye Color Given Hair Color")
```



Note: Mosaic plots are important and reveal useful information especially when we move to comparing more than two proportions, or modeling the relationship between two categorical variables (both concepts will be studied more in depth in later modules)

- **Inference for Difference in Two Proportions**

We continue with an explanation of the R commands and functions needed to conduct the analyses on population proportions. In addition we also provide information about how to use R for multinomial distributions, although this part is optional in Module 2, which focuses primarily on comparing two populations.

We will use here the doctor's survey data from the lecture notes . We will start by reading in the data and setting the order of the categories for each variable to (Yes, No).

```
#Segment 10

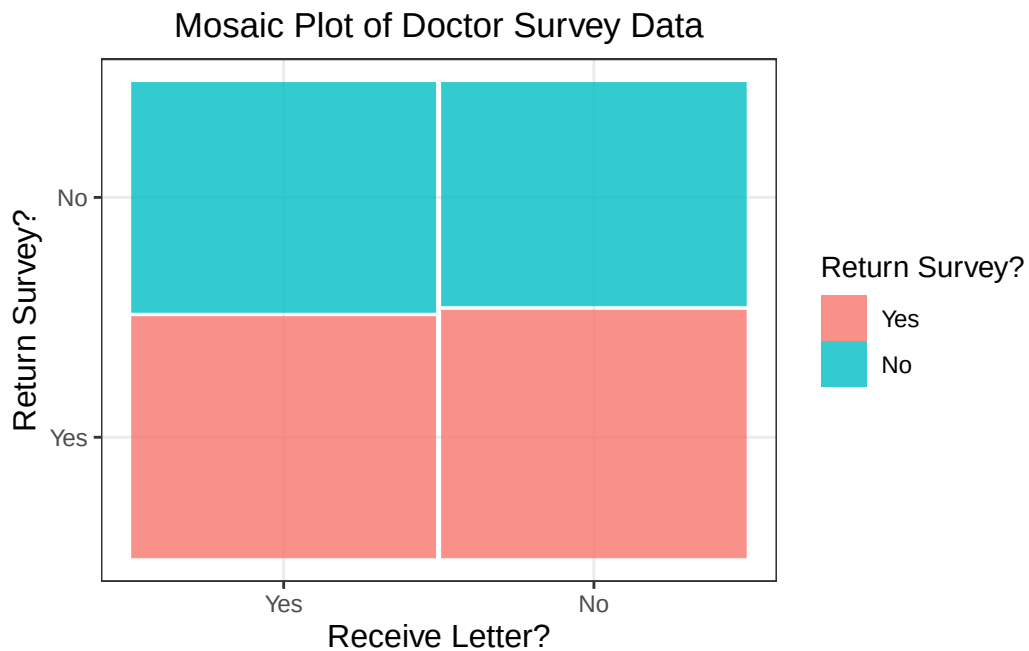
survey.data<- read.csv("m2_s10.doctorsurvey.csv", header = T)
survey.data$Receive.Letter<- factor(survey.data$Receive.Letter,
                                   levels = c("Yes", "No"))
survey.data$Return.Survey<- factor(survey.data$Return.Survey,
                                   levels = c("Yes", "No"))
```

We will then obtain a contingency table for the two variables along with a mosaic plot.

```
survey.table<- table(survey.data$Receive.Letter, survey.data$return.Survey)
survey.table

ggplot(data = survey.data)+
  geom_mosaic(aes(x = product(Receive.Letter),
                      fill = Return.Survey), na.rm = TRUE)+
  theme_bw()+
  theme(plot.title = element_text(hjust=0.5,
                                   size = rel(1.2)),
        axis.title.y = element_text(size = rel(1.1)),
        axis.title.x = element_text(size = rel(1.1)),
        strip.text.y = element_text(size = rel(1.1)))+
  labs(x = "Receive Letter?",
       y = "Return Survey?",
       fill = "Return Survey?",
       title = "Mosaic Plot of Doctor Survey Data")
```

	Yes	No
Yes	2570	2448
No	2645	2384



The number of surveys returned from the letter and non-letter groups are in the first column of the contingency table. We also need to obtain the total number of observations in each group, which is the row margin of the contingency table.

```
groupsuccess<- survey.table[,1]
groupsize<- margin.table(survey.table, 1)
```

To obtain a hypothesis test for the equality of two proportions in R, the function is `prop.test`. You will need to provide the number of successes y and sample size $\{n\}$ for the two groups, along with an alternative (either `two.sided`, `greater`, or `less`). Here is the general command:

```
prop.test(y, n, alternative = "two.sided", correct = F)
```

For example, for the study of the effectiveness of a letter of introduction on response rates for a survey of doctors, the command is:

```
prop.test(groupsuccess, groupsize, alternative = "two.sided", correct = F)
```

2-sample test for equality of proportions without continuity correction

```
data: groupsuccess out of groupsize
X-squared = 1.9143, df = 1, p-value = 0.1665
alternative hypothesis: two.sided
95 percent confidence interval:
 -0.033330958  0.005744447
sample estimates:
   prop 1    prop 2 
0.5121562 0.5259495
```

The value of the z test statistic from class is the square root of the **X-squared** value given in the output. The sign of the z test statistic is the same as the sign of the quantity $\hat{p}_1 - \hat{p}_2$.

Confidence intervals for the difference in two proportions can be handled in a similar manner. First, we will read in the data and set the order of the categories for both variables to (Yes, No).

```
#Segment 12

surgery.data<- read.csv("m2_s11.cancersurgery.csv", header = T)
surgery.data$Surgery<- factor(surgery.data$Surgery, levels = c("Yes", "No"))
surgery.data$Died<- factor(surgery.data$Died, levels = c("Yes", "No"))
```

We will then obtain the contingency table for the two variables along with a mosaic plot.

```
surgery.table<- table(surgery.data$Surgery, surgery.data$Died)
surgery.table

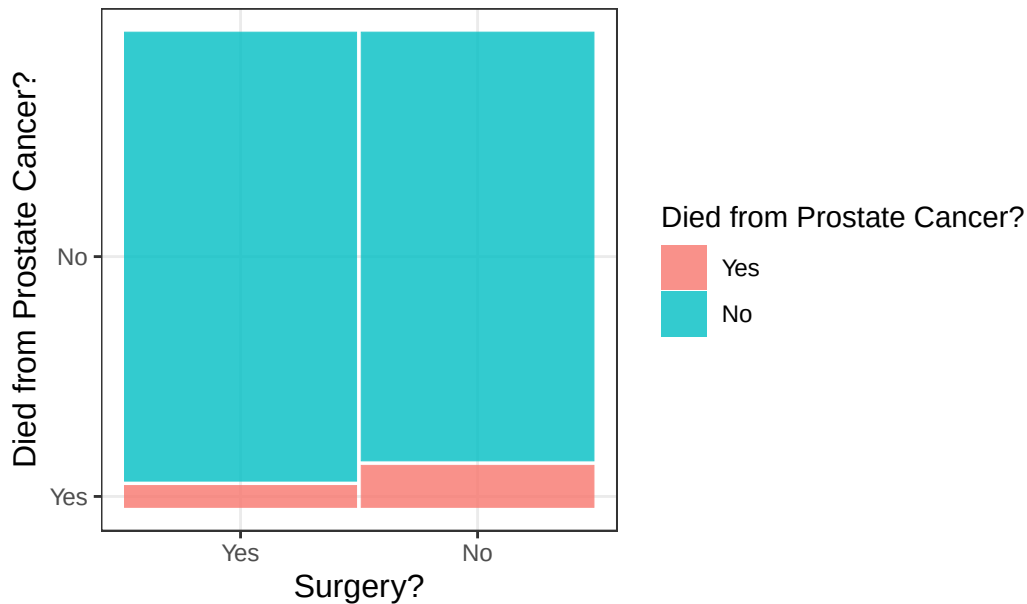
ggplot(data = surgery.data)+
  geom_mosaic(aes(x = product(Surgery),
                    fill = Died), na.rm = TRUE)+
  theme_bw()+
  theme(plot.title = element_text(hjust=0.5,
                                   size = rel(1.2)),
        axis.title.y = element_text(size = rel(1.1)),
        axis.title.x = element_text(size = rel(1.1)),
        strip.text.y = element_text(size = rel(1.1)))+
  labs(x = "Surgery?",
       y = "Died from Prostate Cancer?",
       fill = "Died from Prostate Cancer?",
       title = "Mosaic Plot of Prostate Cancer Surgery Data")

OddsRatio(surgery.table, conf.level = 0.95)
```

	Yes	No	
Yes	16	331	
No	31	317	

odds ratio	lwr.ci	upr.ci
0.4942988	0.2651931	0.9213337

Mosaic Plot of Prostate Cancer Surgery Data



We continue with an example on how to use R to conduct a test of independence for two categorical variables. We will use the example in the lecture notes for the smoking data.

We start by reading in the data,

```
#Segment 13  
smoking.data<- read.csv("m2_s13.smoking.csv", header = T)
```

setting the category order for both variables:

```
smoking.data$Parent<- factor(smoking.data$Parent,  
                             levels = c("Neither", "One", "Both"))  
smoking.data$Student<- factor(smoking.data$Student,  
                              levels = c("Non-Smoker", "Smoker"))
```

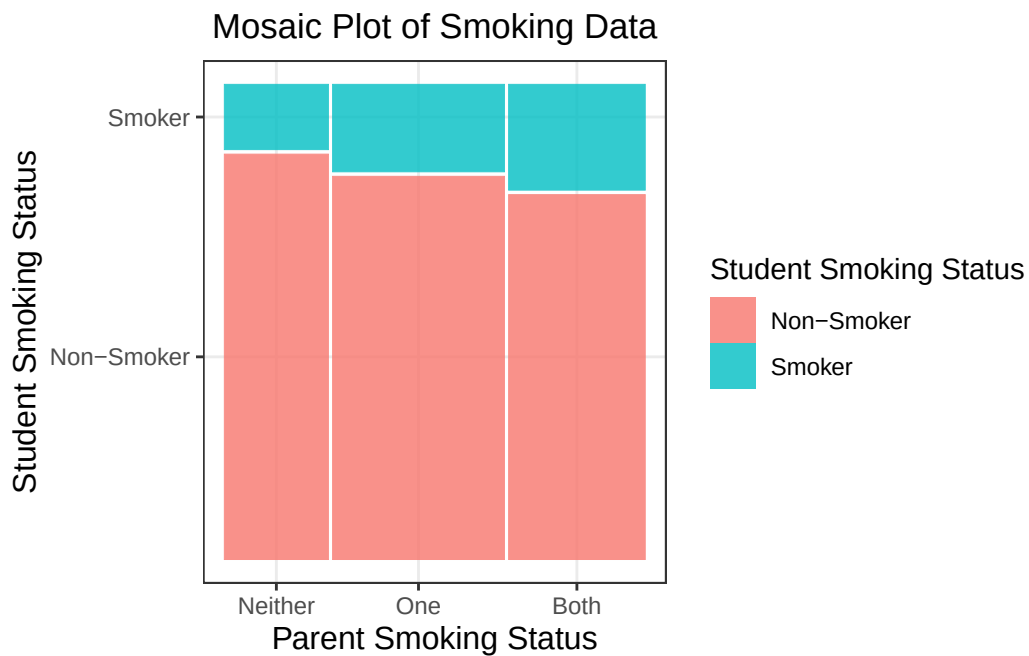
computing the contingency table:

```
smoking.table<- table(smoking.data$Parent, smoking.data$Student)  
smoking.table
```

	Non-Smoker	Smoker
Neither	1168	188
One	1823	416
Both	1380	400

and creating the mosaic plot for the two variables:

```
ggplot(data = smoking.data)+
  geom_mosaic(aes(x = product(Parent),
                      fill = Student), na.rm = TRUE)+
  theme_bw()+
  theme(plot.title = element_text(hjust=0.5,
                                   size = rel(1.2)),
        axis.title.y = element_text(size = rel(1.1)),
        axis.title.x = element_text(size = rel(1.1)),
        strip.text.y = element_text(size = rel(1.1)))+
  labs(x = "Parent Smoking Status",
       y = "Student Smoking Status",
       fill = "Student Smoking Status",
       title = "Mosaic Plot of Smoking Data")
```



To perform the test of independence, we will use the R function `chisq.test` for the chi-square test of independence:

```
smoking.test<- chisq.test(smoking.table, correct = F)
smoking.test
```

Pearson's Chi-squared test

```
data: smoking.table
X-squared = 37.566, df = 2, p-value = 6.959e-09
```

The expected values in the contingency table and the contribution of each cell in the table are saved as the variables

```
smoking.test$expected
(smoking.test$residuals)^2
```

	Non-Smoker	Smoker
Neither	1102.712	253.2882
One	1820.776	418.2244
Both	1447.513	332.4874

	Non-Smoker	Smoker
Neither	3.86551335	16.82884348
One	0.00271743	0.01183057
Both	3.14881241	13.70862455

The following code is given in its most general form, and allows us to compare more than two proportions, by using the `chisq.test()` function in the same way as before:

```
#Segment 14

homesdata<- read.csv("m2_s14.homesdata.csv", header = T)
homesdata$Home.Style<- factor(homesdata$Home.Style,
                             levels = c("Colonial", "Cabin", "A-frame"))
homesdata$Repurchase<- factor(homesdata$Repurchase,
                             levels = c("Yes", "No", "Maybe"))

homes<- table(homesdata)
homes
```

```

ggplot(data = homesdata)+
  geom_mosaic(aes(x = product(Home.Style),
                  fill = Repurchase), na.rm = TRUE)+
  theme_bw()+
  theme(plot.title = element_text(hjust=0.5,
                                  size = rel(1.2)),
        axis.title.y = element_text(size = rel(1.1)),
        axis.title.x = element_text(size = rel(1.1)),
        strip.text.y = element_text(size = rel(1.1)))+
  labs(x = "Home Style",
       y = "Would Repurchase Same Style",
       fill = "Would Repurchase Same Style",
       title = "Mosaic Plot of Home Satisfaction Data")

homes.test<- chisq.test(homes)
homes.test
homes.test$expected

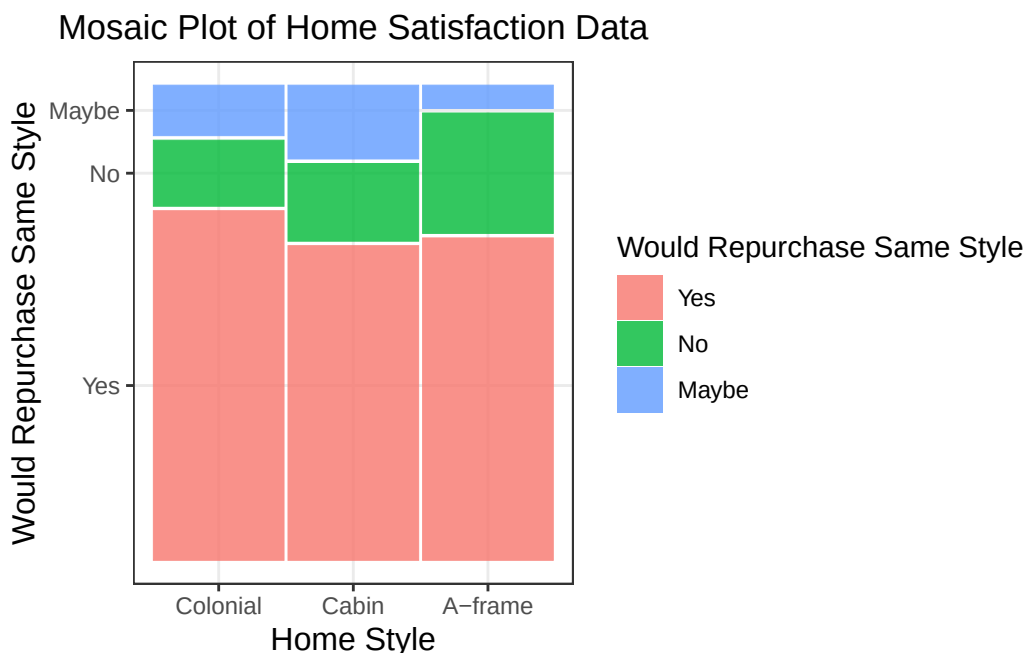
```

	Repurchase		
Home.Style	Yes	No	Maybe
Colonial	90	17	13
Cabin	81	20	19
A-frame	83	31	6

Pearson's Chi-squared test

data: homes
X-squared = 12.006, df = 4, p-value = 0.01731

	Repurchase		
Home.Style	Yes	No	Maybe
Colonial	84.66667	22.66667	12.66667
Cabin	84.66667	22.66667	12.66667
A-frame	84.66667	22.66667	12.66667



The final part of this module is to learn how to use R to calculate McNemar's test for 2 by 2 tables.

McNemar's Test

The base package in R has a function called `mcnemar.test` which will calculate the test statistic and p -value for McNemar's test of marginal homogeneity. To obtain the test statistic and p -value from the lecture notes for the equality of the proportion of correct responses from the two histogram questions, we will first read in the data from the file `histogramq.csv`.

```
#Segment 15

hist.data<- read.csv("m2_s15.histogramq.csv", header = T)
hist.data$Question1<- factor(hist.data$Question1,
                             levels = c("Correct", "Incorrect"))
hist.data$Question2<- factor(hist.data$Question2,
                             levels = c("Correct", "Incorrect"))
```

Then we will calculate the contingency table for the responses from the two histogram questions.

```
hist.table<- table(hist.data$Question1, hist.data$Question2)
```

Finally, we will use the `mcnemar.test` function to obtain the test statistic and p -value for the test. Similar to earlier in this module, we will not use the continuity correction to calculate the test statistic.

```
mcnemar.test(hist.table, correct = F)
```

McNemar's Chi-squared test

```
data: hist.table
```

```
McNemar's chi-squared = 7.7143, df = 1, p-value = 0.005479
```

If the test determines evidence of a difference between the two marginal probabilities, a confidence interval can be calculated using the function `mcnemar.ci`. For the histogram example, a 95% confidence interval for the difference in the proportion of correct responses for the two histogram questions is given by:

```
mcnemar.ci<- function(table, conf.level = 0.95){
  alpha<- 1 - conf.level
  z<- qnorm(1 - alpha/2)

  n<- sum(table)
  y1.<- margin.table(table, 1)[1]
  y.1<- margin.table(table, 2)[1]
  diff12<- table[1,2]-table[2,1]
  add12<- table[1,2]+table[2,1]
  se<- sqrt(add12 - diff12^2/n)/n
  lowerci<- y1./n - y.1/n - z*se
  upperci<- y1./n - y.1/n + z*se
  cat("Confidence Interval = ", lowerci, upperci, "\n")
}

mcnemar.ci(hist.table, conf.level = 0.95)
```

```
Confidence Interval = -0.05198398 -0.009136565
```

#3. Self-Assessment

3.1. Non parametric t -tools: Use the data in `tomato.txt` to answer the following questions. As a reminder, the two variables are treatment (a = new fertilizer, b = control) and yield (lbs of tomatoes).

- a) Draw a side-by-side box plot of the observations in the two treatment groups. Your answer is the plot.
- b) What feature(s) of the box plot suggest that a non-parametric test may be more appropriate than a T-test? Remember that most software shows unusual values individually in the box plot. Answer: Both treatments include an unusually large value. Also, it really looks like the two groups have different variances
- c) Use a Wilcoxon rank-sum test to test the null hypothesis that the two groups have the same median yield. Use an exact p -value calculation, not the normal approximation. Report your p -value. Answer: $p = 0.053$

3.2 Use the creativity data set from the lecture notes to reproduce the numerical answers provided within the notes.

3.3. Comparing proportions

The data given below and repeated in `perry.csv` are from a randomized social experiment started in 1962. 123 three- and four-year old children were randomly assigned to either 2 years of preschool instruction or a control group that received no preschool education. The response, recorded for each individual, was whether or not that individual had been arrested for any crime before the age of 19 (Yes or No). The numbers of individuals in each combination of treatment and response groups are:

	Yes	No
Preschool	19	42
Control	32	30

Note: Know how to do parts 1, 2, 3, and 4 by hand, even if you check your work using the computer.

1. Estimate the proportion of the preschool children who were arrested before age 19.
2. Estimate the standard error of this proportion
3. Estimate the odds that a preschool child will be arrested before age 19.
4. Estimate the odds ratio that fills in the blank in this sentence: The odds that a control child is arrested is _____ times as large as the odds for a preschool child.
5. Estimate a 95% confidence interval for the odds ratio in part 4.
6. Use a Chi-square test to test the hypothesis that control and preschool children are equally likely to be arrested by age 19. Do not use the continuity correction. Report the Chi-square statistic, p -value, and a one-sentence conclusion.

Answers for 3.3:

1. 0.31 or 31% *Note:* 31 (without the %) is not correct, because the proportion is not 31. 31% is numerically = 0.31, which is correct.
2. 0.059 or 5.9% *Note:* calculated as $\sqrt{((0.315(1-0.315))/61)}$, where we carried one additional digit through the intermediate calculation. The sample size (N) is the number of individuals used to estimate this proportion. That's 61, not the total number of individuals in the study (123).
3. 0.46 *Note:* computed as $0.315 / (1-0.315)$, carrying one additional digit through intermediate computations. This question asked for the odds for one group, not the odds ratio that compares groups.
4. 2.36 (or 2.4) *Notes:* the odds of arrest in the control kids is 1.066, so the odds ratio (odds of arrest in control vs preschool) is $1.066 / 0.45 = 2.4$ If you reported circa 0.42, you computed the odds ratio the wrong way around: either odds of wrong outcome (not arrest) or odds of arrest in preschool vs control.
Software makes an arbitrary decision about how to report the odds ratio. The lecture emphasized looking at the two proportions to make sure you have the correct direction. The probability of arrest is larger in the control group so the odds ratio for the wording in the sentence must be > 1 .
5. (1.13, 4.9)
Note: If you reported (0.20, 0.88), you estimated the odds in the wrong direction. Many software will do this. If the odds are 2.4, then the ci for the odds must include that. If software reported (0.20, 0.88), then the answer is the reciprocal of each endpoint.
Additional note: To calculate this without using software, you would calculate the se of the log odds ratio, then the 95% CI for the log odds ratio, then exponentiate those endpoints to get the 95% CI for the odds ratio. This is not something you will be required to do, but is important to know what goes beyond the reported answers.
6. Chi-square statistic = 5.31, $p = 0.021$, Evidence that the preschool intervention reduced the probability of arrest. *Notes:* A causal conclusion is appropriate because this a randomized experiment The p -value here is computed without the continuity correction. If you got $p=0.027$, you included the continuity correction, which is only slightly different from the requested value.